14 October 2026

Dear Editors,

Please consider the enclosed manuscript, "The Norm Blind Spot of Amplitude-Encoding Quantum Kernels in Anomaly Detection", for publication in Quantum Machine Intelligence.

Amplitude encoding is widely used to load classical data into quantum kernels. We show that every kernel machine built on it computes a Rayleigh quotient of a learned matrix. Such models see the direction of an input and never its norm. We test what this means for anomaly detection on four security datasets, with every model trained on the same data at every sample size. The bias causes a clear failure on unseen attack types in simulated data, and a one-amplitude norm anchor repairs it. On two public benchmarks the failure does not appear. We also compare an eight-qubit ZZ feature map with tuned bandwidth and simulate finite-shot kernel estimation.

We think the paper suits the journal because it gives an exact account of an encoding's inductive bias and tests it carefully against classical kernels. That kind of benchmarking is increasingly asked for in quantum machine learning.

The manuscript is original, has not been published, and is not under consideration elsewhere. Code and data preparation scripts are public at https://github.com/arnavk23/quasar-security.

Sincerely,
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